Symmetries and Conservation HW-4

Problem 1. Calculate the matrix $e^{i\alpha A}$, where α is a real number, and

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Solution:
We firstly consider the Jordan canonical form of the matrix.The characteristic polynomial of A is:

$$P(\lambda) = \lambda^3 - \lambda = 0 \implies \lambda_1 = 0, \lambda_2 = -1, \lambda_3 = 1$$

the eigenvectors are:

$$v_1 = (0,1,0)^T, v_2 = (\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2})^T, v_3 = (\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2})^T$$

Then the diagonized A is:

$$A = U \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} U^T$$

where

$$U = [v_1 \ v_2 \ v_3] = \begin{pmatrix} 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 1 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

Then

$$e^{i\alpha A} = U e^{i\alpha D} U^T = U \begin{pmatrix} 1 & 0 & 0 \\ 0 & e^{-i\alpha} & 0 \\ 0 & 0 & e^{i\alpha} \end{pmatrix} U^T = \begin{pmatrix} e^{-i\alpha} & 0 & 0 \\ 0 & \frac{1}{2}(1+e^{i\alpha}) & \frac{1}{2}(1-e^{i\alpha}) \\ 0 & \frac{1}{2}(1-e^{i\alpha}) & \frac{1}{2}(1+e^{i\alpha}) \end{pmatrix}$$

Problem 2. If two matrices satisfy $[A,B] = B$, express $e^{i\alpha A} B e^{-i\alpha A}$ as the combination of $A, B, \ldots$

Solution:
Let $U(\alpha) = e^{i\alpha A} B e^{-i\alpha A}$,then

$$\frac{dU}{d\alpha} = iAU - iUA = i[A,U]$$

Notice that $U(0) = B$,thus by integrating the equation,

$$U(\alpha) = B + i \int_0^\alpha i[A, U(x_1)] dx_1$$

Define the following compressed map:

$$T : X(t) \to B + i\int_0^t [A, X(\tau)]d\tau$$

We see immediately that U is the fixed point of T.Thus by the iteration process:

$$U_1 = B, U_{n+1} = TU_n$$

we have

$$\lim_{n\to\infty} U_n = U$$

Notice that:

$$U_n(t) = B + i\int_0^t dt_1[A, B] + i^2\int_0^t dt_1\int_0^{t_1} dt_2[A, [A, B]] + \ldots + i^{n-1}\int_0^t dt_1 \ldots \int_0^{t_{n-2}} dt_{n-1}[A, [A, \ldots [A, B]]]$$

$$= B + \sum_{k=1}^{n-1} \frac{(it)^k}{k!}([A,)^k, B]$$

Thus,

$$U(\alpha) = B + \sum_{k=1}^{\infty} \frac{(i\alpha)^k}{k!}([A,)^k, B]$$

PROBLEM 3. In class, we have expanded $e^{i\delta_c X_c} = e^{i\alpha_a X_a}e^{i\beta_b X_b}$ to the second order of α or β, and concluded

$$\delta_a = \alpha_a + \beta_a - \frac{1}{2}\alpha_b\beta_c f_{bca}.$$

where f_{abc} is the structure constant of the corresponding Lie algebra. Expand $e^{i\alpha_a X_a}e^{i\beta_b X_b}$ to the third power, and thus express δ_a in terms of α and β up to $O(\alpha^3, \beta^3, \alpha^2\beta, \alpha\beta^2)$.

SOLUTION:

By expanding the RHS to the third order,we have:

$$e^{i\sum_b \alpha_b X_b}e^{i\sum_c \beta_c X_c} = (1 + i\alpha_b X_b - \frac{1}{2}\alpha_b X_b\alpha_d X_d - \frac{i}{6}(\alpha_a X_a)^3)(1 + i\beta_c X_c - \frac{1}{2}\beta_c X_c\beta_d X_d - \frac{i}{6}(\beta_c X_c)^3)$$

$$\Longrightarrow\ e^{i\alpha_b X_b}e^{i\beta_c X_c} = 1 + i\alpha_b X_b + i\beta_c X_c - \alpha_b\beta_c X_b X_c - \frac{1}{2}\alpha_b X_b\alpha_d X_d - \frac{1}{2}\beta_c X_c\beta_d X_d - \frac{i}{2}\alpha_b\alpha[d]\beta_c X_b X_d X_c$$

$$-\frac{i}{2}\alpha_b\beta_c\beta_d X_b X_c X_d - \frac{i}{6}(\alpha_b X_b)^3 - \frac{i}{6}(\beta_c X_c)^3$$

By noticing that

$$-\frac{1}{2}\alpha_b\alpha_d X_b X_d - \alpha_b\beta_c X_b X_c - \frac{1}{2}\beta_c\beta_d X_c X_d = -\frac{1}{2}(\alpha_b X_b + \beta_c X_c)^2 + \frac{\alpha_b\beta_c}{2}[X_b, X_c] = -\frac{1}{2}(\alpha_b X_b + \beta_c X_c)^2 - i\frac{\alpha_b\beta_c}{2}f_{bca}X_a$$

and

$$-\frac{i}{2}\alpha_b\beta_c\beta_d X_b X_c X_d - \frac{i}{6}(\alpha_b X_b)^2 - \frac{i}{6}(\beta_c X_c)^2 - \frac{i}{2}\alpha_b\alpha_d\beta_c X_b X_d X_c$$

$$= -\frac{i}{6}[(\alpha_b X_b + \beta_c X_c)^3 - \alpha_b\alpha_d\beta_c(2X_d[X_b, X_c] + [X_b, X_c]X_d) - \alpha_b\beta_c\beta_d(2[X_b, X_c]X_d + X_d[X_b, X_c])]$$

$$= -\frac{i}{6}(\alpha_b X_b + \beta_c X_c)^3 + \frac{\alpha_b\beta_c}{6}f_{bca}(\alpha_d(2X_d X_a + X_a X_b) + \beta_d(2X_a X_d + X_d X_a))$$

Suppose we make modifications on order 3:

$$\delta_a = \alpha_a + \beta_a - \frac{\alpha_b\beta_c}{2}f_{bca} - g(\alpha^3, \beta^3, \alpha^2\beta, \alpha\beta^2)$$

Then save all terms in LHS to the third order:

$$e^{i\delta_a X_a} = 1 + i\alpha_b X_b + i\beta_c X_c - \frac{1}{2}(\alpha_b X_b + \beta_c X_c)^2 - i\frac{\alpha_b\beta_c}{2}f_{bca}X_a - igX_a - \frac{i}{6}(\alpha_b X_b + \beta_c X_c)^3 + \frac{1}{4}\alpha_d X_d \alpha_b\beta_c f_{bca}X_a$$

$$+\frac{1}{4}\alpha_d\alpha_b\beta_c f_{bca}X_a X_d + \frac{1}{4}\beta_d X_d \alpha_b\beta_c f_{bca}X_a + \frac{1}{4}\beta_d\alpha_b\beta_c f_{bca}X_a X_d + o(\alpha^3, \beta^3, \alpha^2\beta, \alpha\beta^2)$$

By comparing the expansion of RHS and LHS,we see that:

$$igX_a = \frac{\alpha_b\beta_c}{4}f_{bca}(\alpha_d + \beta_d)(X_a X_d + X_d X_a) - \frac{\alpha_b\beta_c}{6}f_{bca}(\alpha_d(2X_d X_a + X_a X_d) + \beta_d(2X_a X_d + X_d X_a))$$

We can thus notice that:

$$igX_a = \alpha_b\beta_c f_{bca}[\frac{1}{4}(\alpha_d + \beta_d)(X_a X_d + X_d X_a) - \frac{1}{6}(\frac{3}{2}(\alpha_d + \beta_d)(X_d X_a + X_a X_d)$$

$$+\frac{1}{2}\alpha_d(X_d X_a - X_a X_d) - \frac{1}{2}\beta_d(X_d X_a - X_a X_d))]$$

$$= \frac{\alpha_b\beta_c f_{bca}}{12}(\alpha_d - \beta_d)[X_a, X_d] = -i\frac{\alpha_b\beta_c(\alpha_d - \beta_d)}{12}f_{bce}f_{eda}X_a$$

Thus,

$$\delta_a = \alpha_a + \beta_a - \frac{\alpha_b\beta_c}{2}f_{bca} + \frac{\alpha_b\beta_c(\alpha_d - \beta_d)}{12}f_{bce}f_{eda}$$

PROBLEM 4. Adjoint representation: In class, we have made the statement that the adjoint reprentation

$$(T_a)_{bc} = -if_{abc}$$

is a representation of Lie algebra. Use Jacobi relation to show that this representation satisfy the commutation rule, that is

$$[T_a, T_b] = if_{abc}T_c$$

SOLUTION:
By direct definition,

$$\forall X \in \mathfrak{g}, [T_a, T_b]X = T_a T_b X - T_b T_a X = [X_a, [X_b, X]] - [X_b, [X_a, X]]$$

By Jacobi Identity,

$$[X_a, [X_b, X]] - [X_b, [X_a, X]] = [X_a, [X_b, X]] + [X_b, [X, X_a]] = -[X, [X_a, X_b]] = [T_a, T_b]X$$

$$\implies [T_a, T_b]X = [X, -if_{abc}X_c] = if_{abc}[X_c, X] = if_{abc}T_c X$$

Thus

$$[T_a, T_b] = if_{abc}T_c$$

$\square$

Problem 5. The $su(2)$ algebra contains three generators $J_a (a = 1, 2,$ and $3)$, that satisfy the commutation rule:

$$[J_a, J_b] = i\epsilon_{abc} J_c$$

Write down its adjoint representation.

Solution:
By definition,

$$(J_a)_{bc} = i\epsilon_{abc}$$

Thus the representation is

$$J_1 \to i(\delta_b^2 \delta_c^3 - \delta_b^3 \delta_c^2)$$
$$J_2 \to i(\delta_b^3 \delta_c^1 - \delta_b^1 \delta_c^3)$$
$$J_3 \to i(\delta_b^1 \delta_c^2 - \delta_b^2 \delta_c^1)$$

If we use matrices,the results are:

$$J_1 \to \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & i \\ 0 & -i & 0 \end{bmatrix}$$

$$J_2 \to \begin{bmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{bmatrix}$$

$$J_3 \to \begin{bmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$